

MATHEMATICAL FORMULATION

F Objective function (multi-objective)

$$\min F = w_1 T_{\text{total}} + w_2 D_{\text{total}} + w_3 C_{\text{total}} + w_4 E_{\text{total}}$$

$T = \text{travel time}$ $D = \text{distance}$ $C = \text{congestion}$ $E = \text{emissions};$
 $\sum w_i = 1$

W Dynamic (traffic-aware) edge weight

$$W_{ij}(t) = \alpha D_{ij} + \beta T_{ij}(t) + \gamma C_{ij}(t)$$

edge cost updates each time traffic conditions change ($\alpha + \beta + \gamma = 1$)

X Decision variable

$$x_{ijk} = 1$$

if vehicle k travels directly from node i to node j

$$x_{ijk} = 0$$

otherwise

C Constraints

1 Vehicle capacity

$$\sum q_i \leq Q_k \text{ for every vehicle } k$$

total demand on a route cannot exceed vehicle capacity

2 Time window

$$a_i \leq \tau_i \leq b_i \text{ for every node } i$$

each stop must be served inside its allowed window

3 Visit exactly once

$$\sum_k \sum_i x_{ijk} = 1 \text{ for every customer } j$$

every customer is served by exactly one vehicle route

4 Depot start / end

$$\sum_j x_{0jk} = \sum_i x_{i0k} = 1$$

every vehicle route begins and ends at the depot

5 Sub-tour elimination

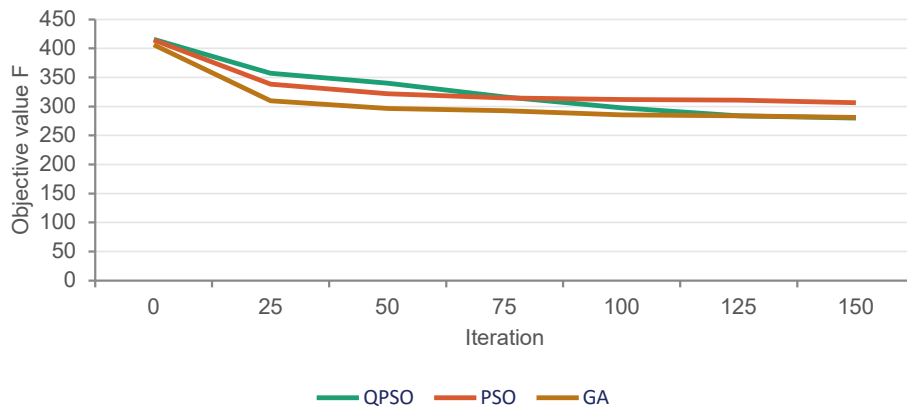
$$u_i - u_j + n \cdot x_{ijk} \leq n - 1$$

prevents disconnected loops that never reach the depot

CONVERGENCE ANALYSIS & BENCHMARKING

1 Convergence curve (fitness vs iteration)

Measured across 12 independent runs — 20-node synthetic instance



Metrics reported (per algorithm, per instance)

Distance

Travel time

Congestion

Runtime

Convergence iter.

Quality vs exact

2 Fair benchmarking protocol

- 1 **Population size** — 30 particles/chromosomes — identical across QPSO, PSO, GA
- 2 **Stopping criterion** — 150 iterations, fixed for every algorithm
- 3 **Independent runs** — 12 runs per instance; report mean $\pm$ std dev
- 4 **Hardware** — Same machine, same session, for every algorithm
- 5 **Problem instances** — Same nodes, demand and time windows for all
- 6 **Random seeds** — Fixed seed set, reused across algorithms

Benchmark hierarchy (progressive scale)

Synthetic
10–50 nodes



Public VRP
benchmark



Real OSM
network



Dynamic SUMO
scenario